

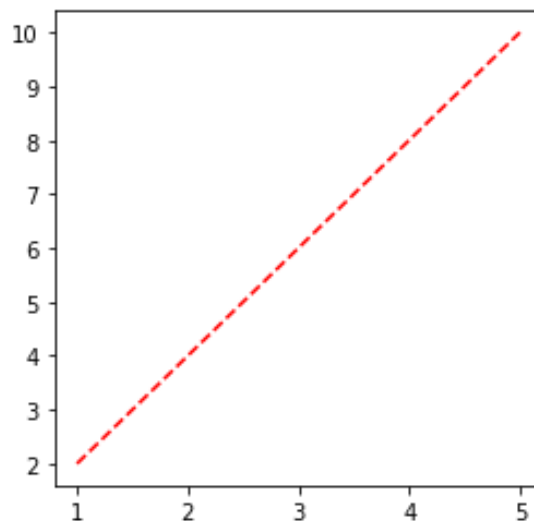
Department: **Data Science and Artificial Intelligence**
Course Name: **Data Science Tools and Software**
Course Code: **03203N**
Professor Name: **Dr. Laila M. Fatehy**
Level: **Two**



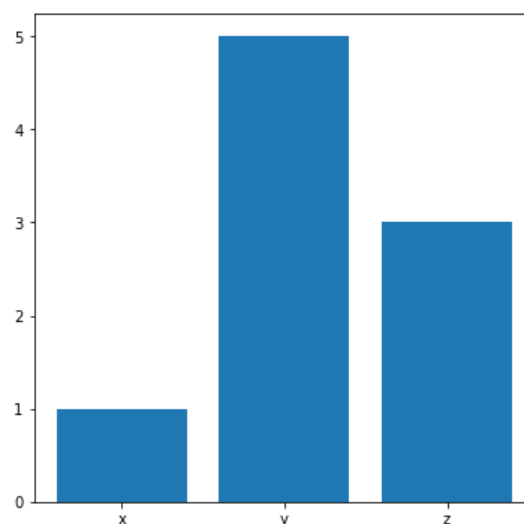
Sheet 6

(Data Visualization)

- (1) Write a Python script to generate a line chart using Matplotlib. The chart should display a red dashed line connecting the data points from the given sets: x-values: (1, 2, 3, 4, 5) , y-values: (2, 4, 6, 8, 10)

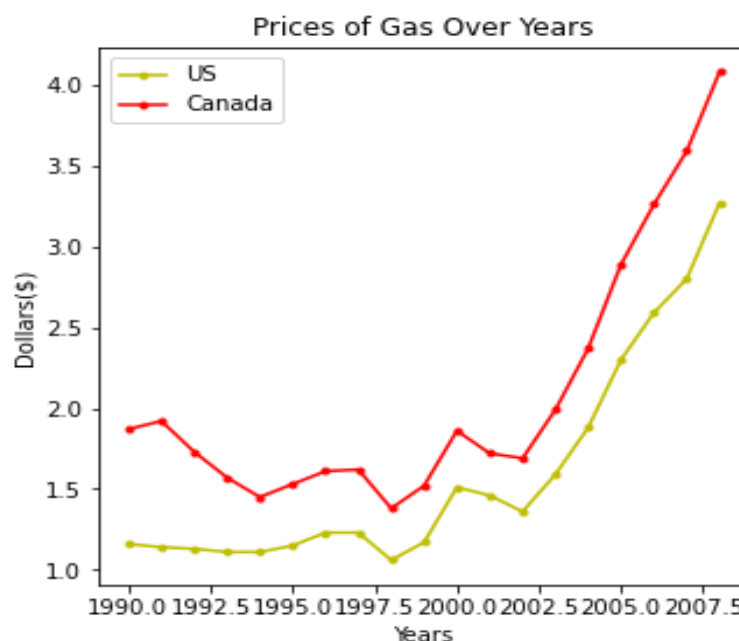


- (2) Develop a Python script to generate a bar chart . The chart should represent three categories labeled x, y, and z, with corresponding values 1, 5, and 3, respectively. Set the figure size to (6,6) and save the generated chart as "bar.png" before displaying it.

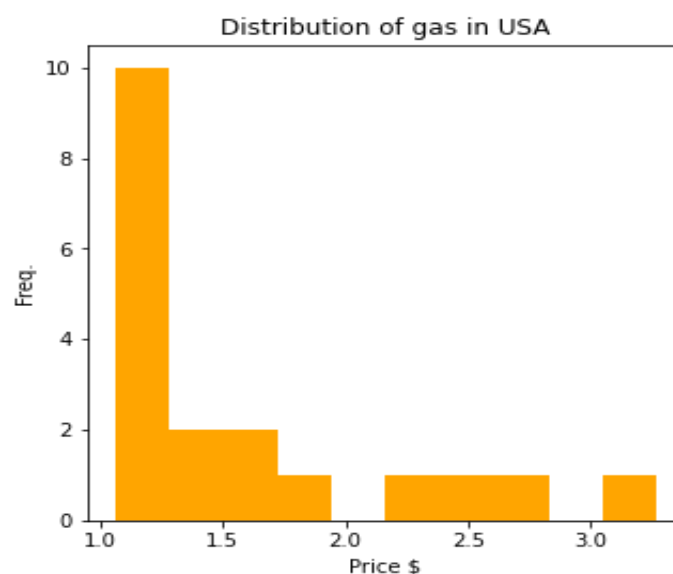


(3) Import a CSV file named "gas_data.csv" using the Pandas library. Load the data into a DataFrame and display its to verify successful import.

(4) Create a line plot that displays two time series on a single figure. The first series represents the prices of gas in the USA over the years and should be plotted in yellow. The second series represents the prices of gas in Canada over the years and should be plotted in red. The figure should have the title "Prices of Gas Over Years", and both the x-axis and y-axis must be labeled.



(5) Write a Python script to create an orange histogram representing the distribution of gas prices in the USA. The histogram should include x-axis and y-axis labels. Additionally, save the generated plot to your computer.

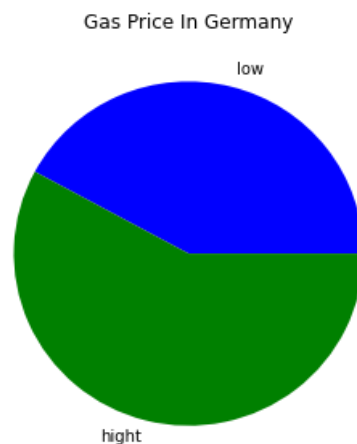


(6) Write a Python script to generate a Pie Chart that visualizes the distribution of gas prices in Germany. The chart should classify gas prices into two categories:

Prices **below 3.5 USD** should be labeled as "**low**" and displayed in **blue**.

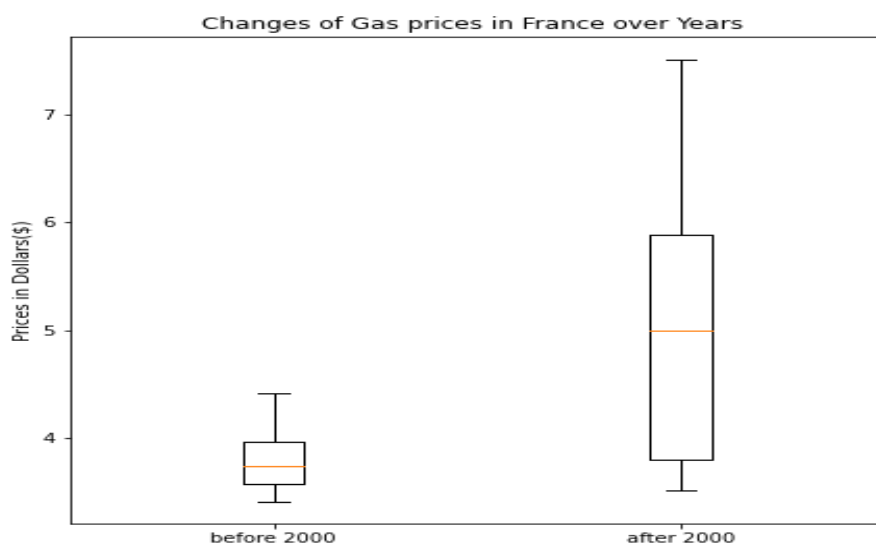
Prices **3.5 USD and above** should be labeled as "**high**", displayed in **green**.

The pie chart should be labeled, titled "Gas Price in Germany", and saved as an image file on the system.

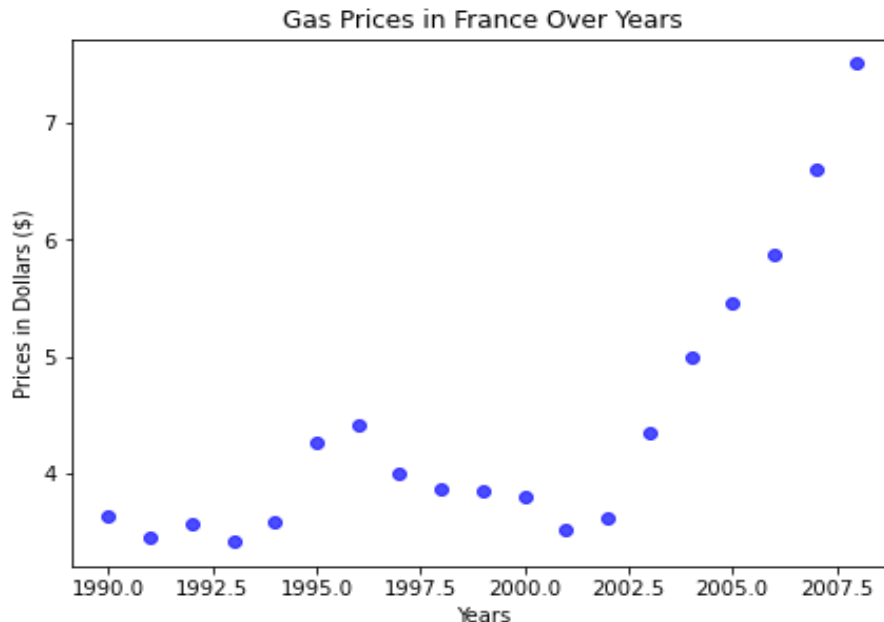


(7) Write a Python script to generate a box plot comparing gas prices in France before and after the year 2000. The figure should contain:

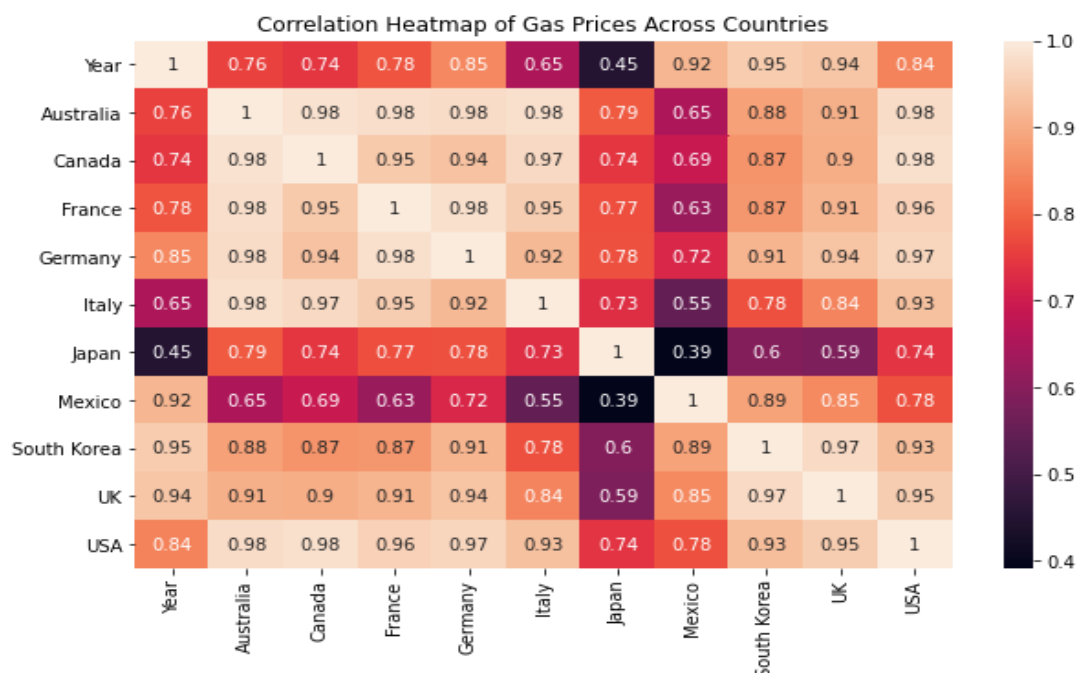
Two box plots in a single figure, representing gas prices before 2000 and after 2000, A y-axis label named "Prices in Dollars (\$)"



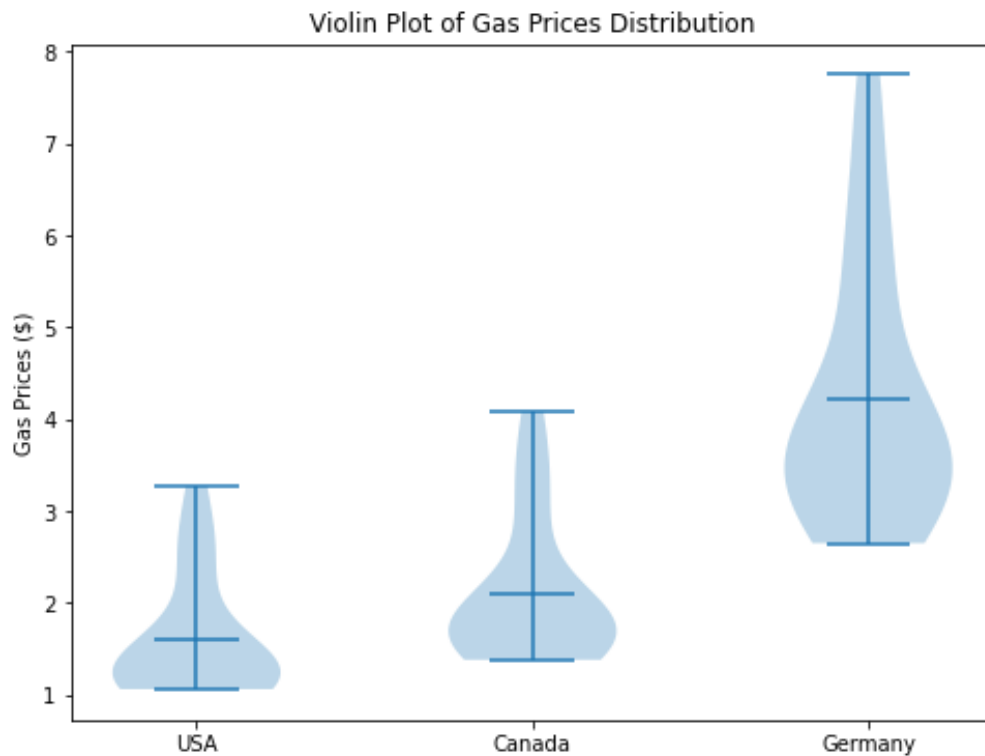
- (8) Write a Python script to generate a scatter plot showing the relationship between gas prices in France and the corresponding years. The figure should include: A scatter plot where the x-axis represents Years and the y-axis represents Gas Prices in France, A title named "Gas Prices in France Over Years".



- (9) Write a Python code to generate a heatmap that visualizes the correlation between different gas prices in various countries from the dataset. The heatmap should display correlation values between columns (countries).



- (10) Create a violin plot to compare the distribution of gas prices in the USA, Canada, and Germany over the years. The plot should show the density of prices for each country.



Good Luck
M. Samy
A. Tarek